Supplementary Material

1 SUPPLEMENTARY TABLES

The Supplementary Material provides extended data and additional analyses that support the findings presented in the main text. Large-format tables summarize the complete incident dataset (2009–2024) and expanded methodological details illustrating cumulative trends, regional distributions, and comparative breakdowns of repeated attack vectors across CEX and DEX platforms. All supplementary tables were created by the authors using curated data extracted from the published literature.

Table S1: High-Impact CEX Exchange Crimes from 2024–2009

DATE	EXCHANGE	PLATFORM	CAUSE OF HACK	LOSS (M$)
31/05/2024	DMM Bitcoin	CEX	Private Key Hack	305
22/06/2024	BtcTurk	CEX	Hot wallet hack	55
18/07/2024	WazirX	CEX	Phishing tactics	230
20/09/2024	BingX	CEX	Security Breach	26
11/09/2024	Indodax	CEX	Hot wallet security breach	22
9/4/23	GDAC	CEX	Obtained access to hot wallet	13
12/11/22	FTX	CEX	Unauthorized transactions	600
1/11/22	Deribit	CEX	Obtained access to hot wallet	28
17/01/2022	Crypto.com	CEX	Unknown	34
9/1/22	LCX	CEX	Obtained access to hot wallet	6.8
11/12/21	AscendEX	CEX	Obtained access to hot wallet	80
5/12/21	BitMart	CEX	Obtained access to hot wallet	150
19/08/2021	Liquid	CEX	Obtained access to hot wallet	97
29/04/2021	Hotbit	CEX	Obtained access to hot wallet	0
1/2/21	Cryptopia	CEX	Hacked	0.045
23/12/2020	Livecoin	CEX	Compromised system/servers	na
21/12/2020	EXMO	CEX	Obtained access to hot wallet	4
1/12/20	BTC Markets	CEX	Internal staff error/mistake	0.270
25/09/2020	KuCoin	CEX	Data leak	275
8/9/20	Eterbase	CEX	Obtained access to hot wallet	5.4
11/7/20	Cashaa	CEX	Malware	3.1
29/06/2020	Balancer	CEX	Vulnerability in protocol	0.5
1/6/20	BuyUcoin	CEX	Data Breach	0.325
14/05/2020	BlockFi	CEX	Data Breach	na
19/04/2020	Lendf.me	CEX	Bugs and Re-entrancy attack	24.5
19/04/2020	Uniswap	CEX	Bugs and Re-entrancy attack	0.5
5/2/20	Altsbit	CEX	Obtained access to hot wallet	0.7
19/12/2019	Youbit	CEX	Obtained access to hot wallet	na

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Table S1 (continued)

DATE	EXCHANGE	PLATFORM	CAUSE OF HACK	LOSS (M$)
26/11/2019	Upbit	CEX	Obtained access to hot wallet	49
5/11/19	Vindax	CEX	Unknown	0.5
11/7/19	Bitpoint	CEX	Compromised system/servers	32
27/06/2019	Bitrue	CEX	Compromised system/servers	4.5
6/6/19	Gatehub	CEX	Unknown	9.5
7/5/19	Binance	CEX	Obtained access to hot wallet	40
29/03/2019	Bithumb	CEX	Suspected trusted insider	29
25/03/2019	Coinbene	CEX	Suspected trusted insider	40
24/03/2019	DragonEX	CEX	Obtained access to hot wallet	1
15/02/2019	Coinmama	CEX	Data leak	0.450
1/2/19	Cryptopia	CEX	Hack and Security Breach	16
26/01/2019	LocalBitcoins	CEX	Phishing data on fake site	0.027
28/10/2018	Maplechange	CEX	Suspected trusted insider	5
14/09/2018	Zaif	CEX	Obtained access to hot wallet	60
9/7/18	Bancor	CEX	Hacked	23.5
18/06/2018	Bithumb	CEX	Hacked	31
10/6/18	Coinrail	CEX	Hacked	40
13/04/2018	CoinSecure	CEX	Suspected trusted insider	3.5
10/2/18	Bitgrail	CEX	Suspected trusted insider	146
27/01/2018	CoinCheck	CEX	Hacked	560
26/01/2018	LocalBitcoins	CEX	Breach six user accounts	0.027
20/12/2017	EtherDelta	CEX	Server DNS compromised	1.4
19/12/2017	Youbit	CEX	Obtained access to hot wallet	na
5/7/17	Bithumb	CEX	Unknown	7
22/04/2017	Yapizon	CEX	Obtained access to hot wallet	5.3
13/10/2016	Bitcurex	CEX	Obtained access to hot wallet	1.5
2/8/16	Bitfinex	CEX	Unknown	623
30/06/2016	ShapeShift	CEX	Obtained access to hot wallet	0.2
9/5/16	Gatecoin	CEX	Obtained access to hot wallet	2.14
7/4/16	Shapeshift	CEX	Suspected trusted insider	0.23
16/02/2016	BTER	CEX	Hack on its cold wallet system	1.75
11/12/15	Bitstamp	CEX	Malware	5.1
15/08/2015	BTER	CEX	Suspected trusted insider	1.65
22/05/2015	Bitfinex	CEX	Obtained access to hot wallet	0.329
18/02/2015	KipCoin	CEX	Wallet servers were hacked	0.728
27/01/2015	796	CEX	Compromised system/servers	0.27
29/07/2014	Cryptsy	CEX	Obtained access to hot wallet	9.58
13/07/2014	Mintpal	CEX	Obtained access to hot wallet	2
11/03/2014	Crypto Rush	CEX	Obtained access to hot wallet	0.63
04/03/2014	Poloniex	CEX	Obtained access to hot wallet	0.05
02/03/2014	Flexcoin	CEX	Obtained access to hot wallet	0.6

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Table S1 (continued)

DATE	EXCHANGE	PLATFORM	CAUSE OF HACK	LOSS (M$)
17/02/2014	Picostocks	CEX	Obtained access to hot wallet	4.434
01/02/2014	Mt. Gox	CEX	Various methods	460
11/11/2013	Bitcash	CEX	Compromised servers	0.1
23/10/2013	Inputs.io	CEX	Bypass 2FA	1.2
10/05/2013	vircutex	CEX	Stolen credentials	0.3
12/09/2012	Bitfloor	CEX	Compromised system/servers	0.25
01/07/2012	Bitcoinica	CEX	Compromised system/servers	0.3
01/05/2012	Bitcoinica	CEX	Compromised system/servers	0.087
01/03/2012	Bitcoinica	CEX	Compromised system/servers	0.228
19/06/2011	Mt. Gox	CEX	Obtained access to hot wallet	8.75
05/10/2011	Bitcoin7	CEX	Obtained access to hot wallet	n/a
			Total	**4,192.5**

Table S2: High-Impact DEX Exchange Crimes from 2024–2009

DATE	EXCHANGE	PLATFORM	CAUSE OF HACK	LOSS (M$)
10/06/2024	UwU Lend	DEX	Hack	19.3
19/04/2024	Hedgey Finance	DEX	Security Breach	44
09/02/2024	PlayDapp	DEX	Private key vulnerability	290
03/09/2024	Penpie Breach	DEX	Market manipulation	27
2/5/23	Level Finance	DEX	security breach	1
28/04/2023	0vix	DEX	flash-loan exploit	2
27/04/2023	Merlin	DEX	HACK	1.82
15/04/2023	Hundred Finance	DEX	HACK	7.4
13/4/2023	Yearn	DEX	exploit	11
9/4/2023	SushiSwap	DEX	Exploit	3.3
1/4/23	AllBridge	DEX	exploit	0.573
29/03/2023	SafeMoon	DEX	Manipulation	9
13/03/2023	Euler Finance	DEX	flash loan attack	200
9/3/23	Hedera	DEX	smart contract exploit	n/a
6/3/23	PeopleDAO	DEX	social engineering hack	0.12
27/02/2023	LaunchZone	DEX	exploit	0.7
17/2/2023	Platypus	DEX	wrong order code	8
13/02/2023	Dforce	DEX	security breach	0
7/2/23	Cow Swap	DEX	hack	0.181
3/2/23	Orion Protocol	DEX	hack	3
12/1/23	BonqDAO	DEX	oracle hack price manipulation	120
12/1/23	LendHub	DEX		6

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Table S2 (continued)

DATE	EXCHANGE	PLATFORM	CAUSE OF HACK	LOSS (M$)
26/12/2022	Defrost Finance	DEX	hack	0
25/12/2022	Rubic	DEX	access to the private keys	1
16/12/2022	Raydium	DEX	exploit	2
11/12/22	Lodestar Finance	DEX	flash loan attack	5.8
2/12/22	Ankr	DEX	exploit	5
11/11/22	DFX Finance	DEX	hack	7.5
2/11/22	Skyward Finance	DEX	exploit	3
1/11/22	Solend	DEX	market manipulation attack	1.26
19/10/2022	Moola Market	DEX	hack	10
11/10/22	TempleDAO	DEX	malicious exploit	2.3
11/10/22	Mango Markets	DEX	exploit and manipulation	100
6/10/22	BNB Chain	DEX	exploit cross-chain bridge	100
2/10/22	Transit Swap	DEX	hacker exploited swap contract	21
8/9/22	New Free DAO	DEX	flash loan attack	1.25
6/9/22	Nereus Finance	DEX	smart contract exploit	0.371
1/9/22	Kyber Network	DEX	vulnerability website exploit	0.265
14/08/2022	Acala	DEX	exploit	n/a
2/8/22	Nomad Bridge	DEX	token bridge exploit	200
24/07/2022	Audius	DEX	malicious governance proposal	6.1
24/07/2022	Horizon Bridge	DEX	exploit	100
16/06/2022	Inverse Finance	DEX	exploit	1.2
1/5/22	Rari Capital	DEX	hack	80
30/04/2022	Saddle Finance	DEX	hack	10
28/04/2022	Deus Finance	DEX	flash loan attack	182
13/04/2022	Elephant Money	DEX	price manipulation attack	11.2
7/4/22	Starstream	DEX	exploit	4
7/4/22	WonderHero	DEX	hack	0.32
2/4/22	Inverse Finance	DEX	exploit	15.6
29/03/2022	Ronin Network	DEX	exploit	625
23/03/2022	Cashio	DEX	exploit	52.8
21/03/2022	One Ring Finance	DEX	hack	1.4
21/03/2022	Li Finance	DEX	smart contract exploit	0.6
20/3/2022	Umbrella Net	DEX	hack	0.7
9/3/22	Fantasm	DEX	hack	2.6
3/3/22	Treasure DAO	DEX	protocol exploit vulnerability	n/a
21/2/2022	Dego Finance	DEX	exploits	10
6/2/22	Meter	DEX	exploits	4.3
3/3/22	Wormhole	DEX	hack	326
3/2/22	KlaySwap	DEX	exploits	1.9

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Table S2 (continued)

DATE	EXCHANGE	PLATFORM	CAUSE OF HACK	LOSS (M$)
28/01/2022	Qubit	DEX	exploits	18.7
1/1/22	Tinyman	DEX	smart contract exploit	3
22/12/2021	Visor Finance	DEX	smart contract exploit	8.1
19/12/2021	Grim Finance	DEX	exploit	30
13/12/2021	Vulcan Forged	DEX	96 private keys were stolen	140
8/12/21	8ight Finance	DEX	private key compromise	1.73
5/12/21	Pizza DeFi	DEX	hack	5
2/12/21	BadgerDAO	DEX	private key compromise	120
30/11/2021	MonoX	DEX	hack	31
5/11/21	bZx	DEX	hack	55
27/10/2021	Cream Finance	DEX	exploits	130
20/10/2021	PancakeHunny	DEX	exploits	2
15/10/2021	Indexed Finance	DEX	exploits	16
30/09/2021	Compound Fin	DEX	exploits	80
21/09/2021	Vee Finance	DEX	exploits	35
20/09/2021	pNetwork	DEX	hack	12.5
16/09/2021	Sushi	DEX	hack	n/a
12/9/21	Zabu Finance	DEX	hack	3.2
4/9/21	Dao Maker	DEX	exploit	4
30/8/2021	Cream Finance	DEX	hack	19
12/8/21	Dao Maker	DEX	hack	7
10/8/21	Poly Network	DEX	hack	3.95
3/8/21	Popsicle Finance	DEX	exploit	25
30/07/2021	Levyathan	DEX	Smart Contract flaw	n/a
23/07/2021	THORChain	DEX	exploit	13
16/07/2021	PancakeBunny	DEX	exploit	2.4
15/07/2021	THORChain	DEX	exploit	4.9
28/07/2021	THORChain	DEX	exploit	0.139
15/07/2021	Bondly Finance	DEX	exploit	n/a
10/7/21	ChainSwap	DEX	hack	4.4
2/7/21	ChainSwap	DEX	hack	0.8
28/06/2021	SafeDollar	DEX	exploit	0.248
22/06/2021	Eleven Finance	DEX	exploit	4.6
21/06/2021	Impossible Fin	DEX	flash loan attack	0.5
16/06/2021	Alchemix	DEX	rugpull	n/a
28/05/2021	Belt Finance	DEX	hack	50
28/05/2021	BurgerSwap	DEX	flash loans	7.2
27/05/2021	Wild Credit	DEX	exploit	n/a
26/05/2021	Merlin Lab	DEX	exploit	0.33
26/05/2021	Merlin Lab	DEX	exploit	0.55
26/05/2021	Merlin Lab	DEX	hack	0.68

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Table S2 (continued)

DATE	EXCHANGE	PLATFORM	CAUSE OF HACK	LOSS (M$)
24/05/2021	AutoShark Fin	DEX	Flash loan attacks	0.822
19/05/2021	Venus Protocol	DEX	price manipulation	n/a
19/05/2021	PancakeBunny	DEX	exploit	200
16/05/2021	bEarn Fi	DEX	exploit	11
12/5/21	xToken	DEX	flash loans exploits	24.5
8/5/21	Rari Capital	DEX	exploit	15
2/5/21	Spartan Protocol	DEX	exploit	30
28/04/2021	Uranium Finance	DEX	platform security incident	50
19/04/2021	EasyFi	DEX	hack	80
4/4/21	Force DAO	DEX	hack	0.376
18/03/2021	TurtleDex	DEX	rugpull	2.4
16/03/2021	Iron Finance	DEX	“subject to an incident”	0.17
14/03/2021	Roll	DEX	exploit	5.7
8/3/21	DODO	DEX	hack	1.9
5/3/21	Paid Network	DEX	contract exploit	160
4/3/21	Meerkat Finance	DEX	hack	31
28/02/2021	Furucombo	DEX	exploit	14
13/02/2021	Alpha Cream Fin	DEX	exploit	37
12/2/21	BT Finance	DEX	exploit	1.5
8/2/21	Growth DeFi	DEX	fake token	1.3
4/2/21	Yearn Finance	DEX	exploit	11
9/1/21	Saddle Finance	DEX	arbitraging for huge profits	0.275
28/12/2020	Cover Protocol	DEX	exploited	5
18/12/2020	Warp Finance	DEX	flash loan	7.7
21/11/2020	Pickle Finance	DEX	hack	30
17/11/2020	Origin Protocol	DEX	re-entrancy attack	7
14/11/2020	Value DeFi	DEX	flash loan	6
12/11/20	Akropolis	DEX	exploit	2
6/10/20	Harvest Finance	DEX	exploiting weak points	24
11/10/20	Leo Finance	DEX	hack	0.042
29/09/2020	Eminence	DEX	smart contract vulnerability	15
13/09/2020	bZx	DEX	faulty smart contracts	8
7/9/20	Soft Yearn	DEX	exploit	0.25
4/8/20	Opyn	DEX	double-spend attack	0.37
29/06/2020	Balancer	DEX	hack that exploited a loophole	0.5
19/04/2020	dForce	DEX	HACK	24.9
15/02/2020	bZx	DEX	Exploit	0.9
11/1/20	Fulcrum	DEX	Flash Loans exploit	2.5
			Total	**4,302.892**

Table S3. Database-ready literature selection queries (Scopus, Web of Science, Google Scholar).

Category	Database-ready query syntax (copy/paste)	Filters
Cybersecurity crimes on cryptocurrency exchanges	• **Scopus:** `TITLE-ABS-KEY(("cryptocurrency exchange*" OR "crypto exchange*") AND (hack* OR breach* OR "cyber attack*" OR "cybersecurity crime*") AND (loss* OR impact OR "case stud*"))` • **Web of Science:** `TS=("cryptocurrency exchange*" OR "crypto exchange*") AND TS=(hack* OR breach* OR "cyber attack*" OR "cybersecurity crime*") AND TS=(loss* OR impact OR "case stud*")` • **Google Scholar:** `"cryptocurrency exchange" (hacks OR breaches OR "cyber attacks") ("financial loss" OR impact) filetype:pdf`	Years: 2009–2024; Language: English; Type: PDF
Emerging quantum threat	• **Scopus:** `TITLE-ABS-KEY(("quantum computing" AND (threat OR risk) AND (cryptocurrency OR blockchain)) OR ("post-quantum cryptography" AND (blockchain OR "crypto exchange*")))` • **Web of Science:** `TS=("quantum computing" AND (threat OR risk) AND (cryptocurrency OR blockchain)) OR TS=("post-quantum cryptography" AND (blockchain OR "crypto exchange*"))` • **Google Scholar:** `"quantum computing" "cryptocurrency security" (threat OR risk) OR "post-quantum cryptography" (blockchain OR "crypto exchange") filetype:pdf`	Years: 2009–2024; Language: English; Type: PDF
Combined cross-topic terms	• **Scopus:** `TITLE-ABS-KEY(("cybersecurity crimes" AND "cryptocurrency exchanges" AND "emerging quantum threat") OR ("crypto exchange hacks" AND "financial loss" AND "quantum computing risk") OR ("high impact cyber attack*" AND "cryptocurrency exchanges" AND "quantum computing") OR ("blockchain security breaches" AND ("quantum threat" OR "quantum cryptography")))` • **Web of Science:** `TS=("cybersecurity crimes" AND "cryptocurrency exchanges" AND "emerging quantum threat") OR TS=("crypto exchange hacks" AND "financial loss" AND "quantum computing risk") OR TS=("high impact cyber attack*" AND "cryptocurrency exchanges" AND "quantum computing") OR TS=("blockchain security breaches" AND ("quantum threat" OR "quantum cryptography"))` • **Google Scholar:** `"cybersecurity crimes" "cryptocurrency exchanges" "emerging quantum threat" OR "crypto exchange hacks" "financial loss" "quantum computing risk" filetype:pdf`	Years: 2009–2024; Language: English; Type: PDF

Google Scholar imposes a ∼256-character limit; split long strings. Replace wildcards per database (e.g., WoS supports ∗). Apply year/language/document filters via each platform's UI.

Table S4. Metadata schema for extracted records

Heading	Definition	Format / controlled values	Example
Authors	First author's surname followed by *et al.* when there are multiple authors.	String: `<Surname> et al.`	Marella *et al.*
Year	Publication year.	Integer (YYYY)	2023
Publication type and outlet category	Type of work and venue class.	*Article*; *Conference paper*; *Preprint*; *Thesis*; *Monograph*; *Report*	Monograph
Area of research / disciplinary domain(s)	Research area; use fixed vocabulary; multiple allowed.	Computer science; Finance; Law; Security (controlled list)	Computer science; Finance
Data type/nature of evidence	What kind of evidence the study uses.	Empirical; Theoretical; Simulation; Case study; Dataset	Empirical (dataset)
Data analysis/analytical methods	Methods applied for analysis.	Descriptive stats; Econometrics / Event study; Time series; ML; NLP; Network analysis	Event study; Random forest
Cryptocurrencies/assets studied and assets explicitly analyzed (name and ticker).	Setting	String: `Name [Ticker]` (list allowed)	Bitcoin [BTC]; Ethereum [ETH]
Crime type (primary)	primary incident/attack class per taxonomy.	Wallet/key compromise; System/server exploit; Smart-contract exploit; Flash loan; Price manipulation; Insider; Data leak	Wallet/key compromise

Table S5. Classical cryptographic mechanisms used by cryptocurrency exchanges

Cryptographic Mechanism	Use in Exchanges	Example	Benefits	Challenges	Reference
Public-Key Encryption	Secure communication, encrypting wallets	User transactions, API security	Confidentiality, secure communication	Private key management is crucial	Subramani et al. (2023)
Hashing Algorithms	Data integrity, password storage	SHA-256 for transactions	Ensures integrity and secure storage	Vulnerable to collision attacks	Banoth and Regar (2023)
Multi-Signature Wallets	Protecting user funds, securing withdrawals	2-of-3 multisig for cold storage	Additional layer of security	Requires coordination among signers	Kiraz (2016)
Digital Signatures	Transaction verification	Transaction signing by the exchange	Authenticity, non-repudiation	Private key management, key loss	Tom et al. (2023)

Table S6. Cryptographic weakness and defense mechanisms in exchanges Badaw et al. (2020); Subramani et al. (2023); Banoth and Regar (2023); Kiraz (2016); Tom et al. (2023)

Cryptographic Issue	Exploited Weaknesses	Defensive Measures	Reference
Weak Encryption	•Deprecated algorithms (e.g., DES, MD5) that are easily breakable. •Low-key strength (e.g., 128-bit AES) vulnerable to brute-force attacks.	•Transition to modern encryption algorithms (e.g., AES-256, RSA-4096). •Regularly update cryptographic standards to phase out deprecated algorithms.	Olaiya et al. (2024); Banoth and Regar (2023)
Poor Implementation of MFA	•SMS-based MFA is vulnerable to SIM-swapping. •Critical actions lack enforced MFA. •Weak or predictable TOTP seeds.	•Use hardware tokens (e.g., YubiKey), or authenticator apps. •Enforce MFA for withdrawals and admin access.	Chenchev et al. (2021); de Saint Guilhem et al. (2020)
Key Management Issues	•Plaintext private key storage. •Single points of failure in key handling. •No key rotation or monitoring.	•Secure keys using HSMs or cold wallets. •Implement rotation policies and access audits.	Homoliak and Perešíni (2024); Gottipati (2020)

Table S7. Security mechanisms on exchanges and quantum vulnerability

Existing Defense Strategies	Techniques	Quantum Vulnerability	Quantum-Era Defense Mechanisms	Ref.
Encryption techniques	Public-key encryption; symmetric encryption; homomorphic encryption	Vulnerable to quantum decryption (e.g., Shor's algorithm)	Lattice-based encryption; hash-based signatures Yang et al. (2023)	Hussain (2023); Jang et al. (2021); Bellamkonda (2019)
Multi-factor authentication (MFA)	2FA; biometrics; hardware tokens	No direct quantum break, but reduced effectiveness if encryption is broken	Continue MFA; integrate with quantum-resistant systems AL-Mubayedh et al. (2019)	Takahashi et al. (2019); Rezaeighaleh and Zou (2020); Prabakaran and Ramachandran (2022); Norta et al. (2022)
Advanced security protocols	Tokenization; HSMs; TLS; VPNs	Potential compromise if underlying crypto is weakened	Post-quantum protocols (e.g., lattice-based KEX, PQ-resilient TLS) Rangan et al. (2022)	Bashir (2020); Aly et al. (2020); Tian et al. (2021)

Table S8: Exchange Platform Architecture: Classical and Quantum Threats and Defense Mechanisms

Exchange Layers	Definition	Classical Threats	Quantum Threats	Defense Mechanisms	Reference
User Interface	The point of interaction for users, including web or mobile applications.	Phishing, Weak authentication, Session hijacking	Quantum phishing, Quantum hacking of weak protocols	Two-Factor Authentication (2FA), User education, Secure socket layer (SSL) encryption	Runsewe et al. (2024); Navamani (2021)
API Layer	Interfaces that allow communication between the exchange platform and third-party services.	Unauthenticated API access, Rate-limiting bypass	Quantum-enabled API exploitations, Insecure token handling	API authentication, Rate-limiting, Secure token-based access	Kamruzzaman et al. (2024); Gupta et al. (2024),Corbet et al. (2019)
Backend Server Layer	Core infrastructure that handles order matching, account management, and risk management.	SQL Injection, Insider threats, Server misconfigurations	Quantum attacks on server encryption, Quantum SQL injection bypass	Intrusion detection systems (IDS), Secure coding practices, Encryption	Cao (2024); Mansour and Kamal (2024), Wilson (2024); Popchev and Radeva (2024)
Blockchain Integration Layer	The system that handles interactions between the exchange and the blockchain network.	Wallet compromise, Blockchain Sybil attacks	Quantum attacks on blockchain consensus algorithms	Post-quantum cryptographic wallets, Quantum-resistant consensus mechanisms	Sion et al. (2024); Jahangir et al. (2024)

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Table S8 (continued)

Exchange Layers	Definition	Classical Threats	Quantum Threats	Defense Mechanisms	Reference
Liquidity & Market Data Layer	Manages liquidity and market data for exchange functionality.	Market manipulation, Fake price feeds, Low liquidity exploitation	Quantum-enabled manipulation attacks on market data	Machine learning-based anomaly detection, Real-time market monitoring	Farooq et al. (2024); Hendrix and Lewis (2021)
Database Layer	The storage system for sensitive information, such as user data and transaction history.	Data breaches, Ransomware, Unauthorized access	Quantum decryption of stored data, Quantum-enhanced ransomware	End-to-end encryption, Regular security audits, Zero-trust architecture	Li et al. (2024); Singamaneni and Muhammad (2024)
Transaction Settlement Layer	Firms record transactions in the blockchain or internal ledger.	Double-spending, Transaction malleability	Quantum double-spending attacks, 51% attacks on PoW systems	Advanced cryptographic algorithms, Blockchain hardening	Alamsyah and Muhammad (2024); Moussa (2020); Sayeed and Marco-Gisbert (2019)
Security Layer	Security mechanisms designed to protect the platform, such as encryption and fraud detection.	Weak encryption, Lack of real-time monitoring	Quantum decryption of encryption algorithms, Quantum-powered fraud detection bypass	Use of quantum-safe encryption algorithms, Real-time monitoring with AI and machine learning	Doris and Gracias (2024); Bucko et al. (2015)

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Table S8 (continued)

Exchange Layers	Definition	Classical Threats	Quantum Threats	Defense Mechanisms	Reference
Compliance & Regulatory Layer	Systems and processes to ensure compliance with regulations, such as KYC and AML.	KYC data breaches, Regulatory non-compliance	Quantum decryption of KYC data	Strong encryption of sensitive data, Adherence to evolving regulations, Continuous monitoring	Vivinvarshan and Priya (2024); Kawai et al. (2023)

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